

Asit Kumar

Software Engineer I · Backend & AI Applications

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PROFESSIONAL SUMMARY

Software Engineer I building backend services and AI-powered applications with Python, FastAPI, PostgreSQL, and Google Agent Development Kit (ADK). Work includes REST APIs, authentication, access control, service integrations, and Retrieval-Augmented Generation (RAG) pipelines spanning document processing, chunking, embeddings, vector retrieval, and LLM integration.

WORK EXPERIENCE

Knowledge ExcelDelhi, India

Software Engineer ISep 2026 – Present

Full-time · Remote

- Build and maintain REST APIs with Python and FastAPI, covering authentication, access control, PostgreSQL data flows, and service integrations.
- Develop agent-based workflows with Google ADK and connect them to backend services and application logic.
- Work on RAG pipelines, from document processing and chunking to embeddings, vector retrieval, and LLM integration.

Software Engineer TraineeMar 2026 – Sep 2026

Internship · Hybrid

- Built FastAPI endpoints and worked on authentication, token refresh, and session management.
- Worked with PostgreSQL queries and AWS S3 integrations, and contributed to an initial RAG retrieval pipeline.
- Contributed changes through Git, code reviews, and the team's deployment workflow.

Planetcast Media Services LimitedNoida, Uttar Pradesh, India

Software Engineer InternOct 2025 – Jan 2026

Internship · On-site

PROJECTS

Smart Food Waste Reduction Platform

Visit project GitHub

Backend system for a donation platform connecting food donors with NGOs in real time.

- Designed and built the backend using Node.js, Express.js, and MongoDB, with geospatial indexing to support location-based donor-to-NGO matching.
- Implemented JWT authentication and RBAC for multi-role access: donors, NGOs, and admins.
- Built a matching algorithm combining distance scoring, urgency level, and NGO capacity to rank donation-to-recipient pairings.
- Used Socket.io for real-time donation status updates and Redis caching to reduce repeated database load on high-traffic endpoints.

Tech stack: Node.js, Express.js, MongoDB, Redis, Socket.io, JWT, RBAC, OpenStreetMap API

EchoSphere — Real-Time Chat Application

Visit project GitHub

Backend for a real-time messaging platform with secure authentication.

- Built an event-driven messaging backend using Node.js and Socket.io to handle real-time message delivery.
- Implemented JWT authentication, OAuth 2.0, and bcrypt password hashing for secure user access.
- Designed RESTful APIs with Express.js middleware, including rate limiting to protect against abuse.

Tech stack: Node.js, Express.js, MongoDB, Socket.io, JWT, OAuth 2.0, bcrypt

CODING PROFILES

- LeetCode:** 200+ problems solved across Data Structures & Algorithms · Contest rating: 1458
- GeeksforGeeks:** 120+ problems solved · Coding score: 310+
- HackerRank:** Certified — Problem Solving (Intermediate), Python (Basic)

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, TypeScript, SQL

Backend: FastAPI, Node.js, Express.js, RESTful APIs, MVC Architecture

Data & caching: PostgreSQL, MongoDB, MySQL, Redis, Database Design, Query Optimization, Indexing

Authentication: JWT, OAuth 2.0, RBAC, bcrypt

Cloud & DevOps: AWS S3, Docker, Git, GitHub, Linux, CI/CD Pipelines, Postman

AI & GenAI: Google ADK, Agent Workflows, RAG, Document Processing, Chunking, Embeddings, Vector Retrieval, LLM Integration

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University (AKTU)2022 – 2026

Bachelor of Technology in Computer Science and Engineering · CGPA: 7.7/10.0